Institute of Technology of Cambodia
Faculty of Electrical Engineering
Génie Informatique et Communication

GICI5CC · CLOUD COMPUTING · WEEK 1

# Introduction to Cloud Computing

Definition, characteristics, history, benefits and challenges.



Heng Rathpisey · Department GIC · Academic year 2025–2026

# Course overview

Institute of Technology of Cambodia · Faculty of Electrical Engineering · Génie Informatique et Communication · Academic year 2025–2026

**PROGRAMME**

Engineering Degree in Computer Sciences

**COURSE**

Cloud Computing · GICI5CC

**CREDITS AND TYPE**

2.5 credits · Core course · 1st semester

**PREREQUISITES**

Operating Systems, Systems and Networks Administration, DevOps

**INSTRUCTOR**

Mr. HENG Rathpisey · M.Eng. UGM Indonesia (2020) · AWS Certified Cloud Practitioner

**CONTACT**

rathpisey@itc.edu.kh · Department GIC

# Course description

## A comprehensive introduction to cloud computing principles and practice, centered on *Amazon Web Services (AWS)* and software development in the cloud.

| | | | |
|---|---|---|---|
| Deployment models | Compute and storage | Networking | Databases |
| Cloud-native apps | Microservices | DevOps | Infrastructure as Code |
| Security | Cost optimization | Serverless | AI/ML services |

Through lectures, laboratory exercises and a capstone project, students learn to architect cloud solutions, implement DevOps workflows and use cloud services for enterprise-grade software development.

# Course learning outcomes

At the end of the course, students are able to:

CLO1 · KNOWLEDGE · C4

## Analyze

cloud computing architectures, service models, AWS services and cloud-native design patterns to evaluate and recommend appropriate cloud solutions for diverse application requirements.

CLO2 · SKILL · P5

## Implement

scalable cloud applications using AWS services, DevOps practices, Infrastructure as Code and containerization technologies to deploy production-ready solutions.

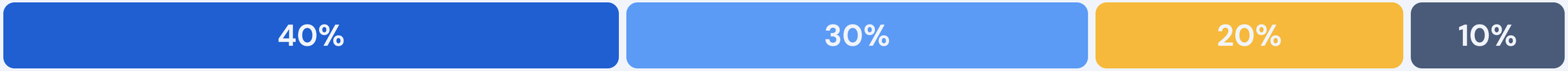
CLO3 · ATTITUDE · A4

## Document

cloud architectures, configurations and implementation decisions through professional lab reports and project documentation, demonstrating clear technical communication and accountability.

ASSESSMENT

# How the course is graded




## 40%

**Final exam**

Closed book · Topics 6–12

## 30%

**Labs and capstone**

Lab reports and the capstone project

## 20%

**Midterm exam**

Closed book · Topics 1–5

## 10%

**Quizzes**

Short quizzes and class participation

# Assessment methods and outcomes

| Method | What it covers | Weight | Outcomes |
|---|---|---|---|
| **Quizzes and participation** | Short quizzes and active class participation, covering all topics | 10% | CLO1, CLO2 |
| **Lab assignments and capstone** | Weekly lab reports and the capstone project, with documentation and presentation | 30% | CLO1, CLO2, CLO3 |
| **Midterm exam** | Closed book. 20 multiple-choice, 10 short-answer and 1 AWS lab exercise. Topics 1–5 | 20% | CLO1 |
| **Final exam** | Closed book. Multiple-choice, short-answer and problem-solving questions. Topics 6–12 | 40% | CLO2 |

**Weight by learning outcome:** CLO1 30% · CLO2 55% · CLO3 15%

# Course policy

## < 30

Overall score below 30 out of 100 means a re-sit of the exam.

## > 50%

Absence without permission over 50% of the total course hours is an automatic fail.

## 3 weeks

No weekly project update for more than 3 weeks means zero for the poster presentation.

# Student responsibilities

1. Create and maintain an AWS Academy Learner Lab account (quota 50 USD per account)
2. Monitor AWS usage to stay within the quota limit
3. Complete weekly lab assignments and submit lab reports on time
4. Work consistently on the capstone project throughout the semester
5. Participate actively in class discussions and demonstrations
6. Document all AWS resources created and clean them up after labs
7. Back up project code and configurations regularly
8. Attend all lectures and lab sessions, and adhere to academic integrity policies

# Course schedule · Weeks 1–8

| Week | Topic | Lab or project |
|---|---|---|
| 1 | Introduction to cloud computing and its evolution | — |
| 2 | Deployment and service models: IaaS, PaaS, SaaS | — |
| 3 | Cloud infrastructure, EC2 and virtualization | Lab 1: AWS account and EC2 |
| 4 | Cloud scaling and load balancing | Lab 2: Auto scaling and ELB |
| 5 | Cloud storage: S3, EBS, EFS and data management | Lab 3: S3 storage |
| 6 | Cloud networking (VPC) and content delivery | Lab 4: VPC and CloudFront |
| 7 | Databases in the cloud: RDS and DynamoDB | Lab 5: RDS and DynamoDB |
| 8 | Open week, no topic listed in the specification | — |

# Course schedule · Weeks 9–16

| Week | Topic | Lab or project |
|---|---|---|
| 9 | Cloud-native applications and microservices | Lab 6: Containers (ECS/EKS) |
| 10 | Security, IAM and compliance in AWS | Lab 7: IAM and security |
| 11 | Cloud cost management and economics | Lab 8: Cost monitoring |
| 12 | DevOps, CI/CD pipelines and Infrastructure as Code | Capstone: planning |
| 13 | AI/ML services in the cloud | Capstone: development |
| 14 | Capstone project work | Capstone: implementation |
| 15 | Review of the whole course | — |
| 16 | Final examination and project submission | — |

# Tools and references

SOFTWARE

## Tools

AWS Console, AWS CLI and AWS Academy Learner Lab

VS Code

Git

Docker

Terraform

Postman and a web browser

READING

## Textbooks

**Required:** AWS Certified Solutions Architect Official Study Guide, 3rd edition. Baron, Baz, Bixler, Gaut et al. Sybex, 2020.

**Recommended:** Cloud Computing: Concepts, Technology and Architecture (Erl, Puttini, Mahmood; Prentice Hall, 2013) and Architecting the Cloud (Kavis; Wiley, 2014).

LEARNING

## Online resources

AWS Documentation

AWS Well-Architected Framework

AWS Whitepapers

AWS Training and Certification

AWS Labs and hands-on tutorials

# Labs

## AWS Academy Cloud Foundation

50% coupon toward the AWS Certified Cloud Practitioner exam.

## AWS Educate

Getting started with compute, networking, security, storage, databases and serverless.

## AWS Academy Learner Lab

Hands-on lab account for the weekly exercises, with a 50 USD quota per account.

## Bonus for AWS Certified students

Students who earn an AWS certification receive bonus credit.

# Week 1 learning objectives

1. Define cloud computing
2. Describe the five essential characteristics of cloud computing
3. Summarize the history and evolution of cloud computing
4. List the benefits of cloud computing
5. Enumerate the challenges of cloud computing

THIS WEEK

## Topic 1

Introduction to cloud computing: concepts and evolution. Assessed by class participation and a quiz (CLO1).

DISCUSSION

# What is the cloud?

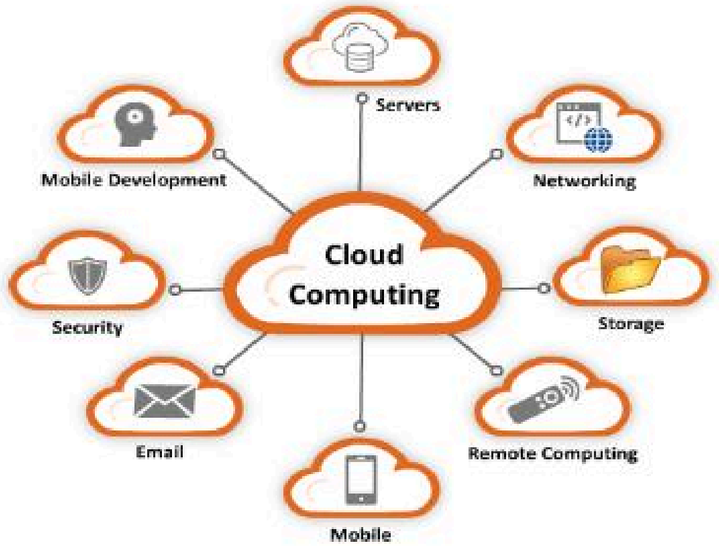
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# A cloud computing analogy



You don't buy a cow for a bottle of milk.



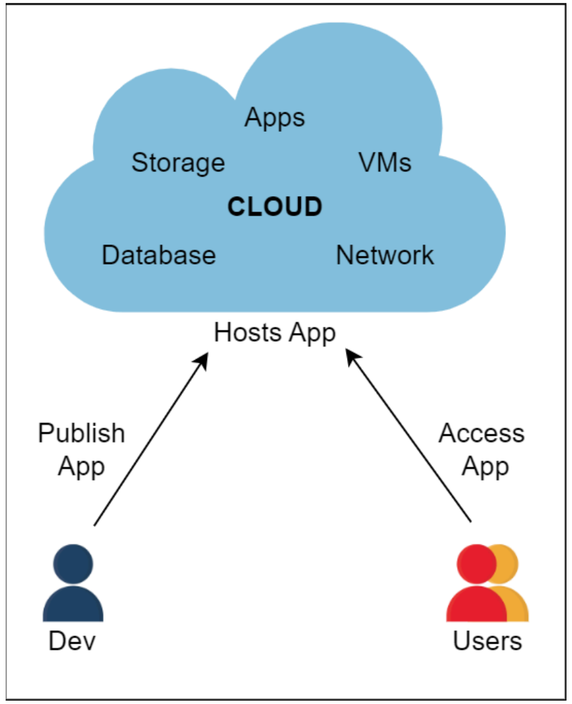

You buy the computer resources that you need instead of the whole package.

You buy the computing resources you need, not the whole package.

# What is cloud computing?

The on-demand delivery of computing services *over the internet*, “the cloud”.

- **Servers**
- **Storage**
- **Databases**
- **Networking**
- **Software**
- **Analytics**



# Three definitions

### Wikipedia

Shared pools of configurable system resources and higher-level services, rapidly provisioned with minimal management effort.

### AWS

On-demand delivery of compute power, database storage, applications and other IT resources over the internet, with pay-as-you-go pricing.

### NIST

Convenient, on-demand network access to a shared pool of configurable computing resources, rapidly provisioned and released with minimal effort.

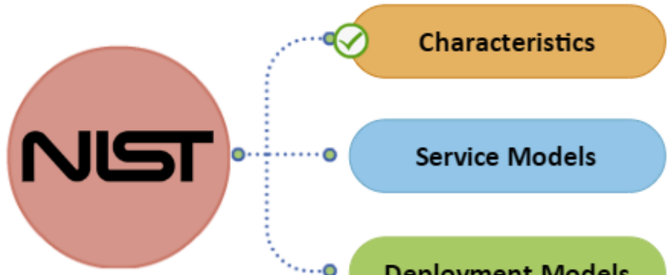
In this course the NIST definition is the standard.

IN PLAIN WORDS

# A cloud is just a bunch of computers in a datacenter.

It is a real physical place, reached over the internet, where a service is performed for you or your stuff is stored. Explain the Cloud like I'm 10, Todd Hoff.

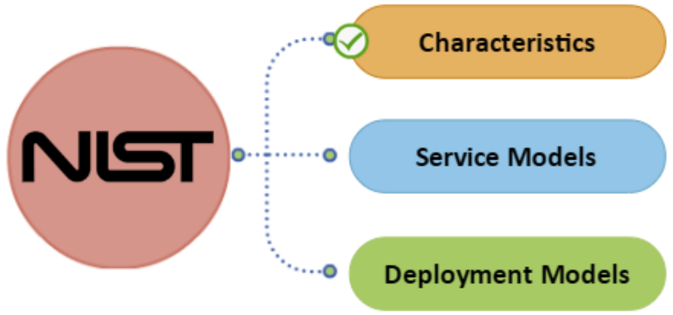
# NIST key takeaways




Characteristics: the five essentials of cloud computing

1. On-demand self-service
2. Broad network access
3. Resource pooling
4. Rapid elasticity
5. Measured service

# NIST service models

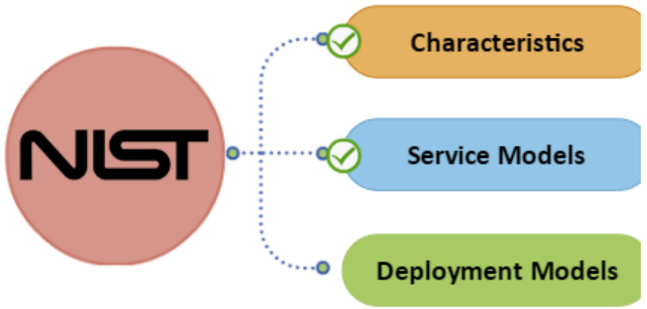



## Software as a Service (SaaS)

The cloud provider's application is hosted on a cloud infrastructure and used by customers across the world.

For example, Gmail is a cloud-hosted application run by Google.

# NIST service models



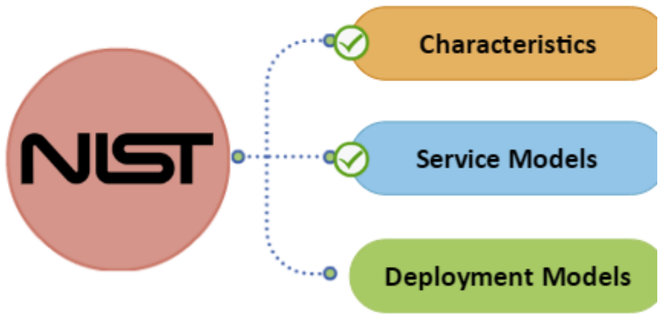

## Platform as a Service (PaaS)

Applications deployed on the cloud infrastructure are offered by the cloud service provider. For example, deploying applications.

## Infrastructure as a Service (IaaS)

The provisioning of fundamental computing resources for application deployment. For example, creating virtual machines on Amazon Web Services or Microsoft Azure.

# NIST deployment models




## Public cloud

Infrastructure for the general public. Government services used by the public are public cloud.

## Private cloud

Infrastructure solely for a single organization, such as applications used in schools or offices.

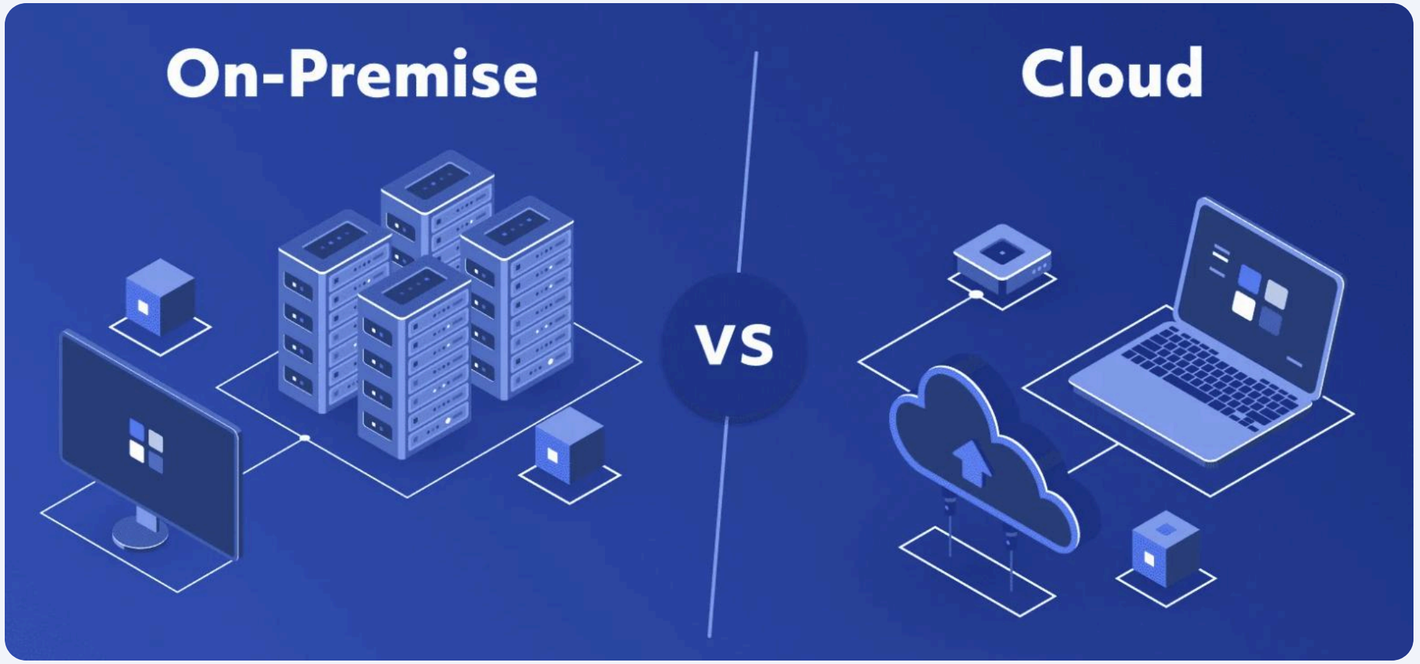
## Community cloud

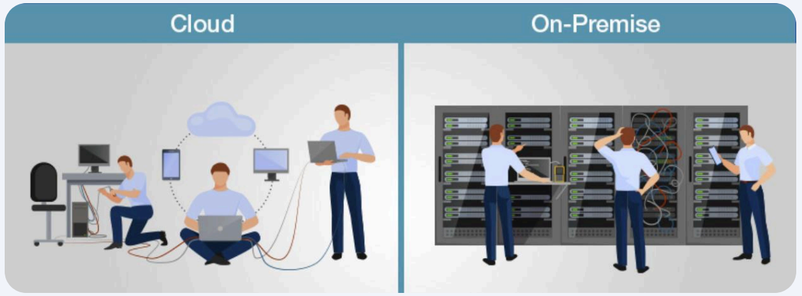
Infrastructure for a community with similar concerns. For example, Facebook.

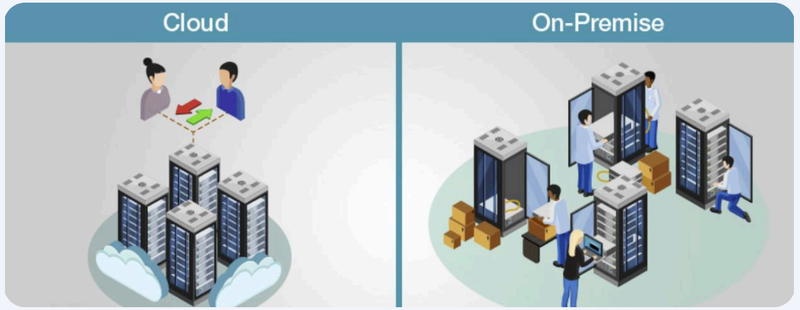
## Hybrid cloud

A combination of two or more clouds, used by an organization or academic institution.

# On-premises vs cloud: why businesses shift



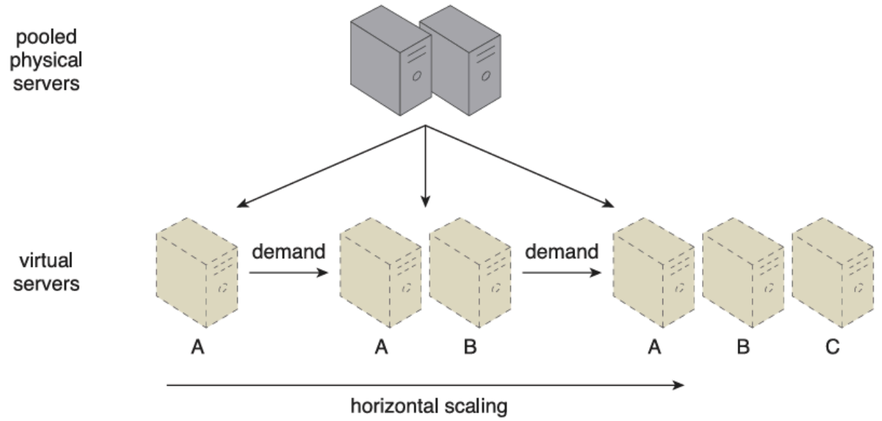



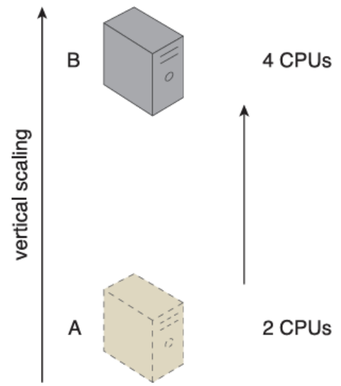

Why do businesses shift from the traditional on-premises model, hosting on local servers, to the cloud computing model?

# Flexibility and scalability

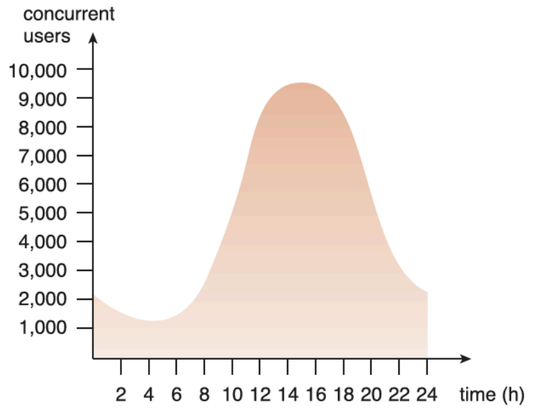
Quickly scale up or down based on demand.




Horizontal scaling: add virtual servers as demand grows




Vertical scaling: more CPUs per server




Concurrent users peak during the day

# Cost efficiency

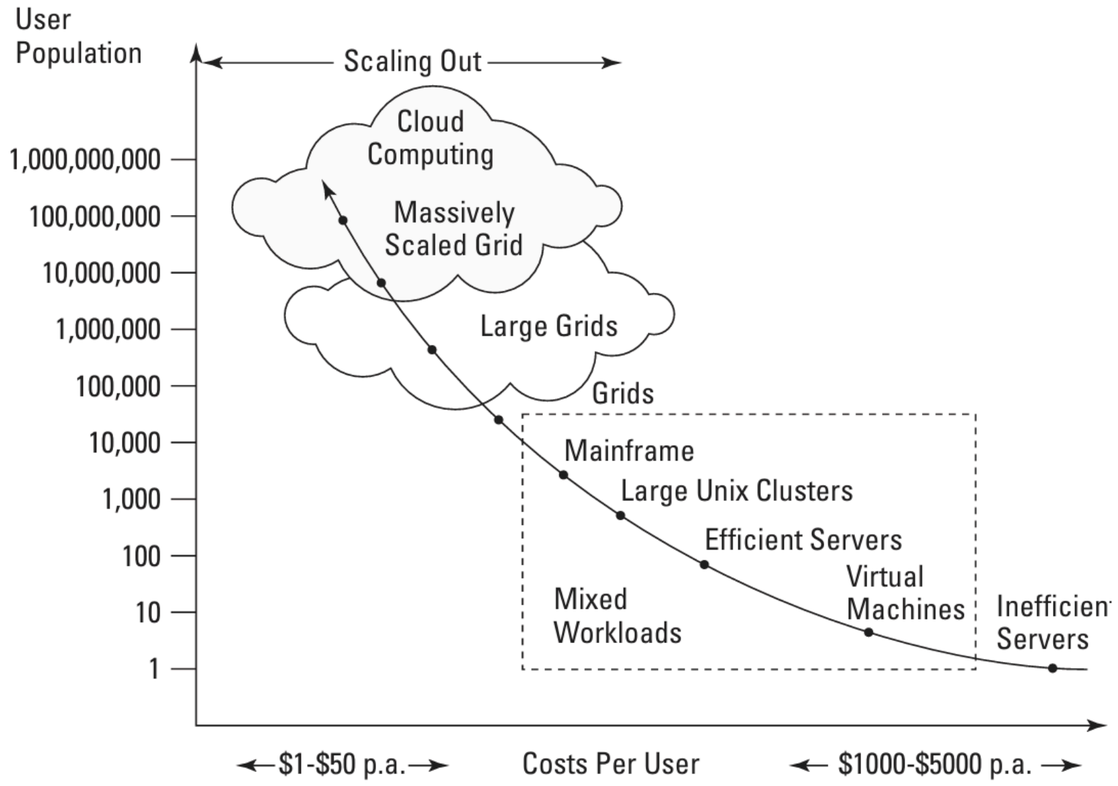
Eliminate the capital expense of buying hardware and running on-site datacenters. Pay for what you use with flexible pricing models.

**$1,000–$5,000**

per user per year with inefficient servers

**$1–$50**

per user per year at cloud scale



# Performance and speed

Providers use the *latest, fast, efficient hardware* and global datacenters to reduce latency.

## Regions

Datacenters are grouped into regions around the world.

## Availability zones

Each region holds several zones, so applications can run close to their users.



# Availability

## Users can customize their applications using cloud services from *anywhere*.

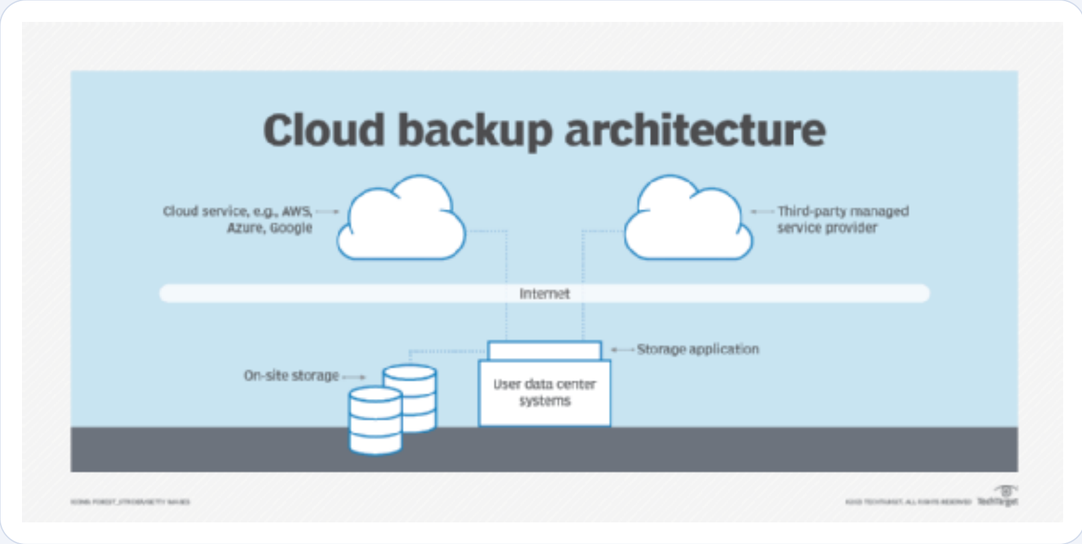
Cloud services reach laptops, tablets, phones and watches over the internet.



# Security




Datacenters are physically secured and run by the provider.




Cloud backup: on-site data is copied to a cloud service or a managed provider over the internet.

# Why businesses move to the cloud

### Flexibility and scalability

Scale up or down quickly as demand changes.

### Cost efficiency

No upfront hardware or on-site datacenter. Pay for what you use.

### Performance and speed

Modern hardware and global datacenters reduce latency.

### Availability

Use and customize your applications from anywhere.

# On-premises vs cloud

| | On-premises | Cloud |
|---|---|---|
| **Upfront cost** | Buy hardware, build a datacenter | No capital expense, pay as you go |
| **Scaling** | Buy and install more servers | Scale up or out on demand |
| **Maintenance** | Your own team runs it | Provider runs the hardware |
| **Access** | Local server | Anywhere over the internet |

# Five essential characteristics

01

### On-Demand Self-Service

Provision and release resources yourself, with no human help from the provider.

02

### Broad Network Access

Reach services over the network from laptops, phones and tablets.

03

### Resource Pooling

Shared physical and virtual resources serve many tenants, assigned by demand.

04

### Rapid Elasticity

Scale out or in quickly, even automatically, so capacity feels unlimited.

05

### Measured Service

Usage is metered and reported, so you pay for what you use.

# Why know these characteristics?

### Self-service

Access resources anytime without waiting.

### Anywhere

Use services from any location.

### Shared

Efficient and cost-effective for many users.

### Scaling

Adjust resources to your needs.

### Pay for use

Only pay for what you actually use.

### A shared language

NIST's definition is the standard reference, so providers, customers and engineers describe the cloud the same way.

### Real cloud or not

If a service lacks self-service or metering, it is only hosting. The five traits are a checklist for spotting real cloud.

### Better decisions

Each trait ties to a benefit and a cost, so you can choose, size and budget cloud services with confidence.

# History of cloud computing

## 1960s

Mainframe timesharing: users rent a time slot to run calculations.

## 1970s

Virtual machines emerge and are rented out with private networks.

## 1990s

The web arrives, servers are virtualized and computing becomes a utility.

## 2006

Amazon launches AWS, lending its spare computing capacity to the market.

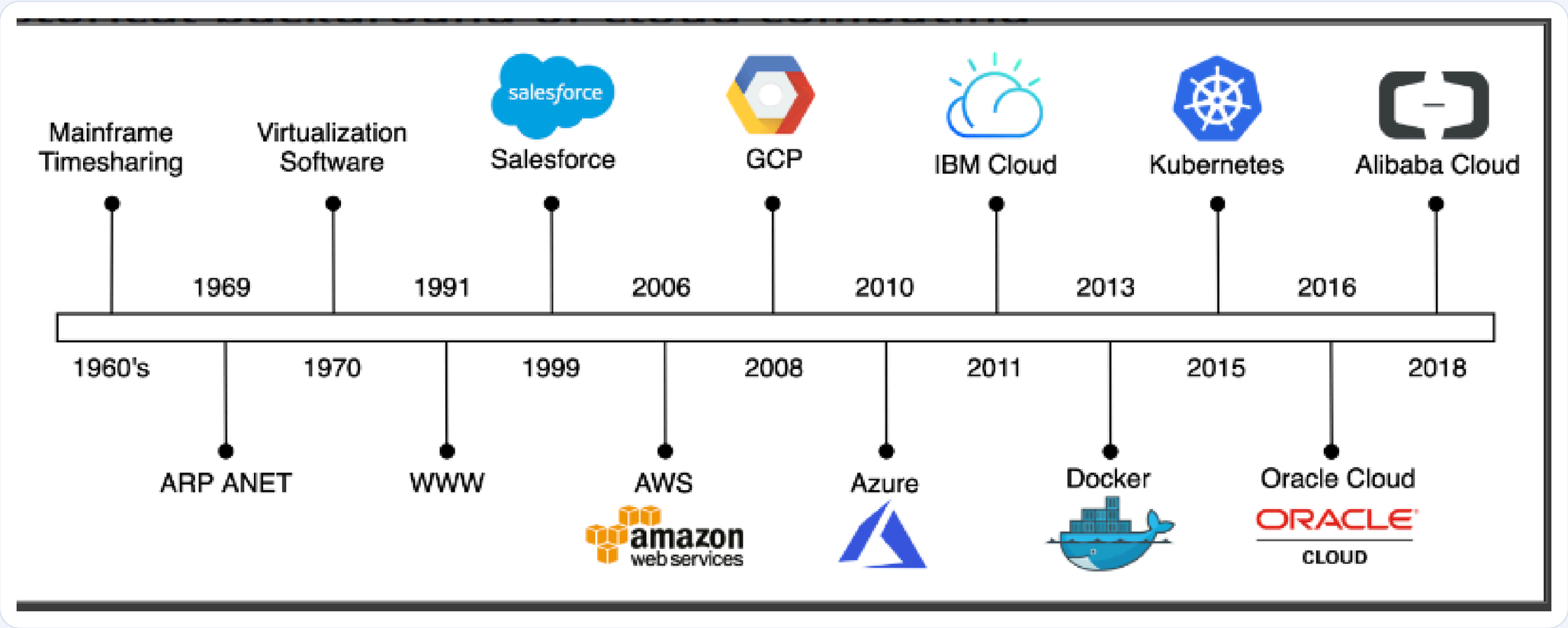
## 2008–10

Google Cloud and Microsoft Azure join the race.

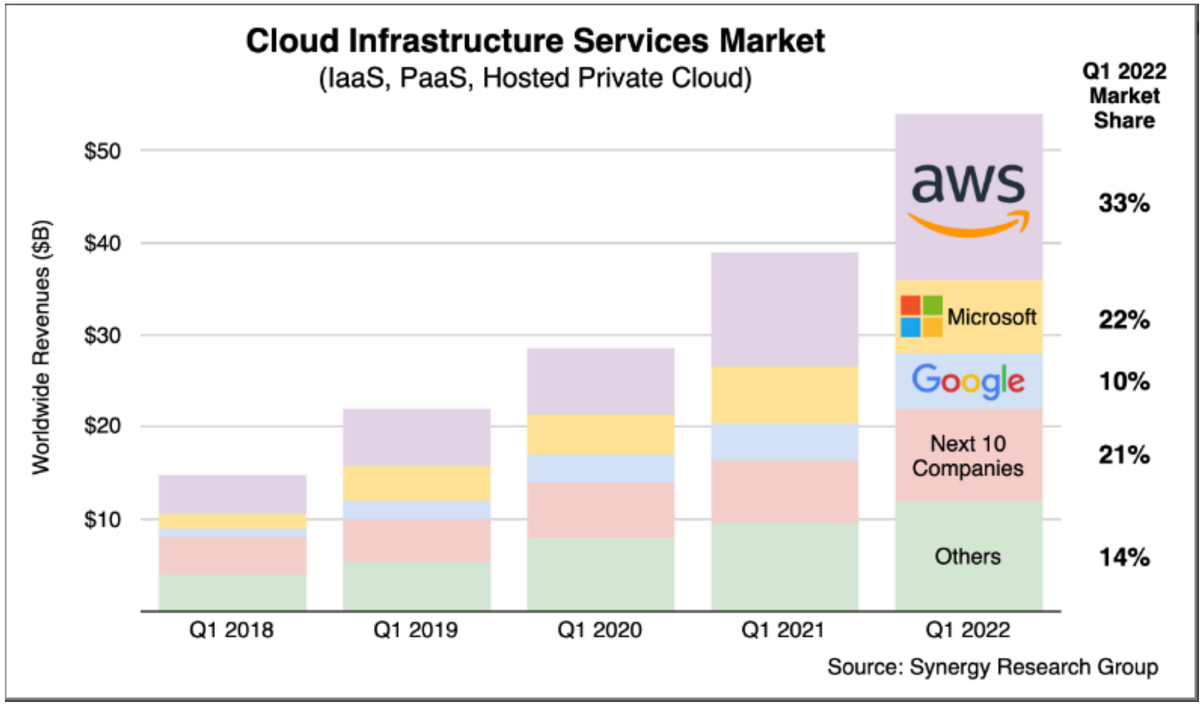
## 2013+

Docker and Kubernetes make containers the cloud-native standard.

# Cloud computing timeline



# Major providers of cloud services




33% **AWS**

22% **Microsoft Azure**

10% **Google Cloud**

# First-tier and second-tier providers



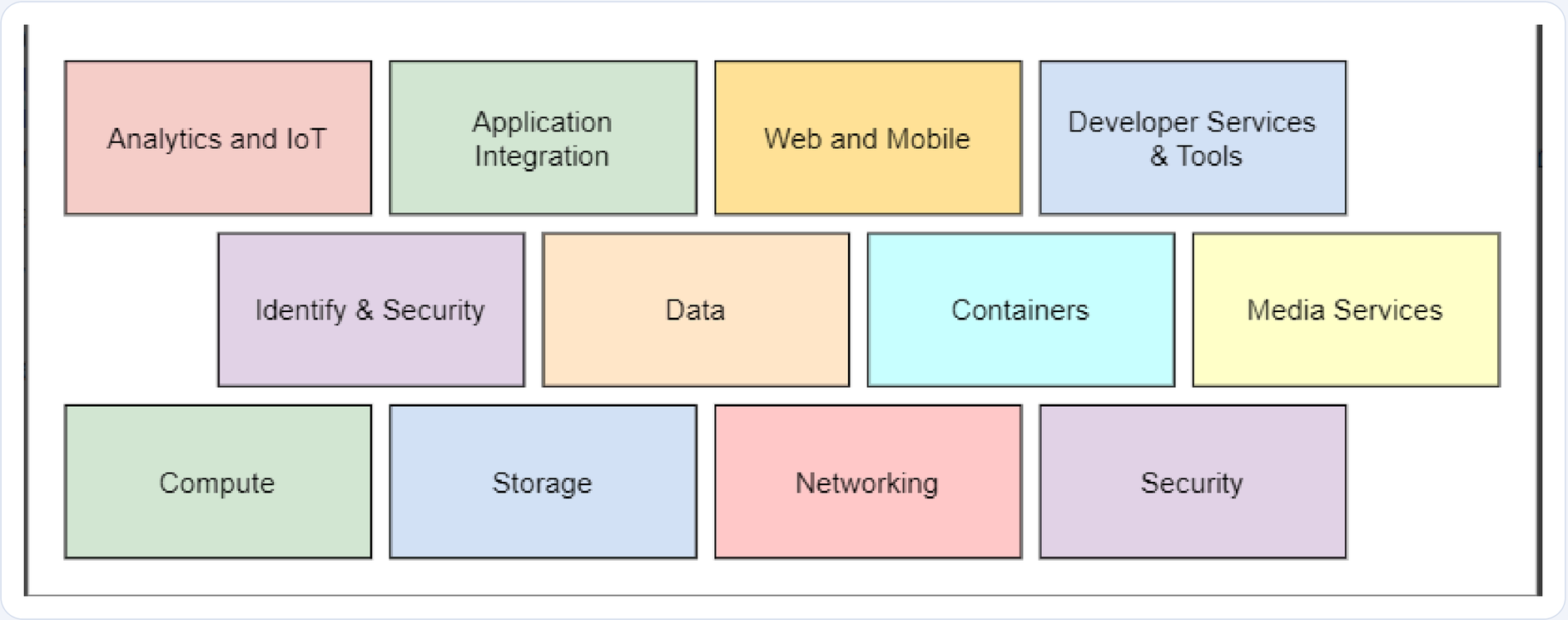

## First tier

AWS · Microsoft Azure · Google Cloud Platform

## Second tier

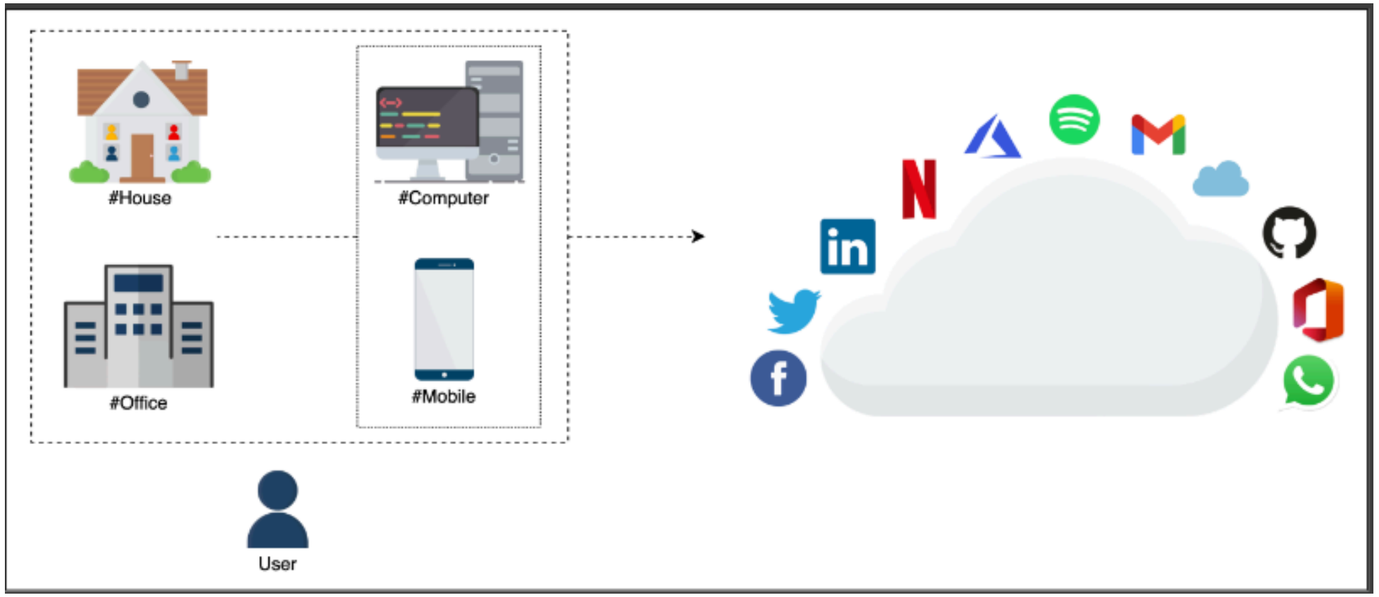
IBM Cloud · Alibaba Cloud · Oracle Cloud · Tencent Cloud · Salesforce · SAP

# Cloud services




There are hundreds of services, and they are similar across platforms: an AWS service usually has a match in Microsoft Azure.

# Where the cloud is used




Most of the services we use every day run on cloud platforms: Netflix, Amazon Prime, Gmail, Spotify, Uber, WhatsApp, iCloud and GitHub.

These billion-dollar companies do not run their own datacenters. They focus on the customer experience and leave running and maintaining services to cloud vendors.

# Largest companies on cloud platforms

## Some of the Largest Companies Using Cloud Platforms

| Company | Number of Active Monthly Users | Cloud Platform |
|---|---|---|
| **Netflix** | 200+ million | AWS |
| **Spotify** | 433 million | GCP |
| **Twitter** | 450+ million | GCP |
| **Instagram** | 2+ billion | AWS |
| **Adobe (Creative Cloud)** | 30+ million | Microsoft Azure / AWS |

### AWS

Netflix and Instagram

### Google Cloud

Spotify and Twitter

### Azure and AWS

Adobe Creative Cloud

# Some use cases of cloud

### Software

Delivery of software and services.

### Streaming

Audio and video.

### Cloud-native

Cloud-native applications.

### Backup

Store, back up and recover data.

### AI

Data analysis, machine learning and AI workloads.

DISCUSSION

# What do you think could be the cons of cloud?

Join at slido.com with the event code **[code]**

# Challenges of cloud computing



## Cost

Running in the cloud is not automatically cheaper. Choosing the right service for the use case is part of the design.



## Security

Protecting sensitive data has been debated since the start, and data breaches remain a real concern.



## Vendor lock-in

Switching providers can cost so much that customers stay even when they are unhappy.

# Why should we use the cloud?

### Developer or software engineer

Working knowledge of the cloud is almost essential; most organizations run, or are migrating, their workloads there.

### IT professional or admin

These roles have shifted heavily toward the cloud, where most hardware and software is managed.

### Executive

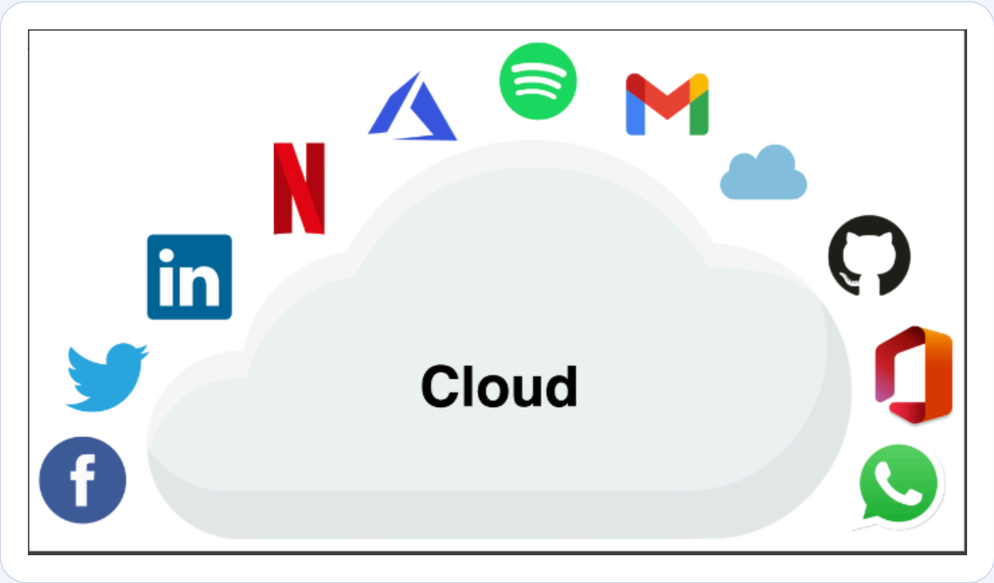
You are part of the digital transformation that almost every business is going through.

### Entrepreneur

One person or a very small team can build a robust business on a global scale.

### Consumer

Everyday apps such as Netflix, Gmail and WhatsApp run on the cloud.



# Emerging technologies powered by cloud



## IoT

Combating rhino poaching in South Africa.



## AI

Digital experiences for millions of United States Tennis Association fans.



## Blockchain

Food supply chain traceability that helps farmers reduce waste.



## Analytics

Predictive maintenance for a city’s infrastructure, by KONE.

# Next session

01

## Common cloud services

02

## Cloud deployment models

03

## Cloud service models